

Experiment 19

Title

Write a Prolog Program for Student–Teacher–Subject Database

Aim

To implement a Student–Teacher–Subject database using Prolog facts and retrieve teacher and student information through queries.

Objective

To understand how Prolog stores facts and uses rules to search and retrieve information from a simple database.

Algorithm

Step 1: Define facts for teachers and the subjects they teach.

Step 2: Define facts for students and the subjects they study.

Step 3: Create a rule to retrieve the subject handled by a teacher.

Step 4: Create another rule to retrieve the students studying a particular subject.

Step 5: Execute the required query and display the matching result.

Program

% Teacher and Subject Database

teacher(john, math).

teacher(jane, english).

teacher(bob, science).

teacher(sue, history).

teacher(tom, art).

% Student and Subject Database

student(alice, math).

student(alice, science).

student(bob, english).

student(bob, science).

student(carol, history).

student(carol, art).

student(dave, math).

student(dave, english).

student(dave, art).

% Retrieve teacher subject

teacher_subject(Teacher, Subject) :-

teacher(Teacher, Subject).

% Retrieve students studying a subject

subject_student(Subject, Student) :-

student(Student, Subject).

Sample Input

?- teacher_subject(john, Subject).

Sample Output

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\--- ---/ ---
| ?- consult('C:/Users/B SANTIHOSH KUMAR/OneDrive/Documents/exp19.pl').
compiling C:/Users/B SANTIHOSH KUMAR/OneDrive/Documents/exp19.pl for byte code...
C:/Users/B SANTIHOSH KUMAR/OneDrive/Documents/exp19.pl compiled, 31 lines read - 1935 bytes written, 9 ms

yes
| ?- teacher_subject(john, Subject).

Subject = math

(15 ms) yes
| ?-

teacher_subject(bob, Subject).

Subject = science

yes
.\
```

Result

Hence, the Prolog program for the Student–Teacher–Subject database was executed successfully and the correct output was obtained.